

Ambisonics Encoding of Room Impulse Responses using a Device-Agnostic Diffusion Model

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Abstract—We address the problem of encoding room impulse responses (RIRs) into high-order Ambisonics (HOA) representations from arbitrary and potentially insufficient or incomplete microphone array measurements. This task is fundamentally ill-posed for microphone arrays with limited spatial capture capabilities, such as irregular or sparse arrays, as classical linear methods fail to reconstruct high-order spatial detail. We introduce a diffusion-based generative framework that models the statistical properties of HOA RIRs. This enables device-agnostic encoding from arbitrary microphone arrays, potentially unseen during data measurement. Our approach incorporates a posterior sampling procedure that enforces consistency between the estimated signals and the measurements while plausibly reconstructing spatial information that is unobservable from the limited measurements alone. Experiments on simulated data demonstrate that our method outperforms linear and neural baselines, achieving accurate HOA RIR estimation up to 12th order. A listening test with binaural renderings, including both simulated and measured RIRs, further confirms that the proposed method yields higher perceptual similarity to reference Ambisonics RIRs than all baselines. The flexibility and accuracy of the proposed framework opens new possibilities for scalable acoustics simulations.

Index Terms—spatial audio, generative models, microphone arrays

I. INTRODUCTION

Ambisonics provides a compact representation of a spherical sound field at the point of observation via Spherical Harmonics Decomposition (SHD), enabling flexible rendering to arbitrary listening / head directions, and playback configurations [1]. When the sound field is represented using higher SHD orders, the Higher-Order Ambisonics (HOA) signal can express finer spatial detail. However, accurate HOA acquisition typically requires dense spherical sampling [2]. Estimating HOA signals from irregular microphone arrays, such as wearable devices, is challenging because the available measurements often lack sufficient spatial bandwidth to resolve higher-order components [2].

This work focuses on Ambisonics encoding of Room Impulse Responses (RIRs) from microphone-array measurements. RIRs encapsulate the acoustic propagation and are central to downstream spatial audio applications, including acoustic simulation, binaural and loudspeaker rendering, and data generation for training and evaluating speech enhancement and spatial audio models. Representing RIRs in the HOA domain is particularly attractive because it disentangles the room acoustics (captured as SHD coefficients) from the specifics of any given recording device: once an HOA RIR is available, the soundfield can be rotated arbitrarily (the SHD is directionally continuous) and sampled/decoded at arbitrary directions. Furthermore, since the SHD is device-agnostic, we may map the SHD RIR to microphone signals of different microphone arrays given their Array Transfer Functions (ATFs), by imprinting new acoustic scattering and sampling properties. This ability is valuable for scalable data generation, cross-device evaluation, and rapid iteration over array designs without repeating time-consuming measurement campaigns.

Classical Ambisonics encoding methods typically implement a linear least-squares projection from microphone signals to SHD coefficients [3]. For irregular microphone arrays, the encoding operator

often becomes poorly conditioned or rank-deficient, limiting the spatial detail that can be reconstructed from the available signals. Prior work has introduced additional structure into the inverse problem through model assumptions or learned priors, including signal sparsity [4] and parametric spatial audio [5]. However, these approaches are primarily designed for perceptual relevance and rely on assumptions that may limit reconstruction accuracy when violated.

Data-driven methods based on deep neural networks have recently been proposed for related tasks. Neural Ambisonics encoders predict time-dependent encoding matrices from specific microphone arrays to Ambisonics formats [6], [7], but are typically limited to first-order and device-specific cases, despite efforts to improve robustness across arrays [8]. Ambisonics order upmixing, such as extrapolating First-Order Ambisonics (FOA) to HOA, has also been explored using parametric spatial audio methods [3] and also predictive neural networks [9] and generative diffusion models [10]. Importantly, both neural approaches have focused on scene-based audio signals, with no demonstrated results for accurate RIR encoding.

In this work, we formulate the Ambisonics encoding of array RIRs as an inverse problem and address it within a Bayesian probabilistic framework. The ill-posed nature of the task motivates the use of generative models, which can sample from learned posterior distributions rather than producing a single deterministic estimate. Diffusion models have recently emerged as a powerful approach [11], serving as expressive priors for solving inverse problems [12]–[15]. We propose a diffusion model for HOA RIRs based on a hybrid architecture that operates jointly in the time–frequency and time domains. Building on this model, we introduce an iterative inference procedure based on diffusion posterior sampling that leverages the device-agnostic generative prior to solve the inverse problem for any array with a known ATF. Experiments in simulated environments show improved performance over classical least-squares encoding and neural baselines for RIRs up to 12th-order Ambisonics. Finally, we provide evidence of the perceptual relevance of the proposed approach through the results of a formal listening test conducted with both simulated and measured room impulse responses.

II. PROBLEM DEFINITION

We consider a source–receiver configuration in a reverberant enclosure, and a microphone array with M microphones. Let $\mathbf{x} \in \mathbb{R}^{M \times T}$ denote the multichannel RIR measured at the array, where T is the number of time samples. The room response around the receiver unit sphere can be represented as a continuum of plane waves with a directional amplitude density. A compact representation is obtained by expanding this directional density in spherical harmonics up to order N , yielding $(N+1)^2$ time-domain HOA coefficient signals. We collect these coefficients over time in $\mathbf{a} \in \mathbb{R}^{(N+1)^2 \times T}$, where each row corresponds to a spherical-harmonic component $\mathbf{a}_{nm}[t]$ with order $n \in \{0, \dots, N\}$ and degree $m \in \{-n, \dots, n\}$ [2].

Let $\mathbf{h} \in \mathbb{R}^{M \times (N+1)^2 \times T}$ denote the ATF in the time domain, which was obtained on a dense spherical grid of directions and then transformed to the SHD. In practice, $\mathbf{h}$ can be obtained from measurements or simulations, and depends on the array geometry, device scattering, and microphone characteristics.

In the frequency domain, the forward model decouples across frequency bins and simplifies to a matrix multiplication. For each frequency bin $k \in \{0, \dots, K-1\}$,

$$\mathbf{X}[k] = \mathbf{H}[k]\mathbf{A}[k], \quad (1)$$

where $\mathbf{X}[k] \in \mathbb{C}^M$ is the DFT of the measurements $\mathbf{x}$, $\mathbf{H}[k] \in \mathbb{C}^{M \times (N+1)^2}$ is the DFT of the ATF $\mathbf{h}$, and $\mathbf{A}[k] \in \mathbb{C}^{(N+1)^2}$ is the DFT of the HOA coefficients $\mathbf{a}$. For notational simplicity, we omit the frequency index k in subsequent expressions; all matrix multiplications are understood to be carried out independently for each frequency bin. Given the measurements $\mathbf{x}$ and the ATF $\mathbf{h}$, the goal is to estimate the Ambisonics coefficients $\mathbf{a}$.

A standard solution for Ambisonics encoding is the least-squares estimate $\hat{\mathbf{A}}_{\text{LS}} = \mathbf{H}^\dagger \mathbf{X}$ [3], where $\mathbf{H}^\dagger \in \mathbb{C}^{(N+1)^2 \times M}$ denotes the Moore–Penrose pseudoinverse of $\mathbf{H}$ (computed per frequency), and is defined as $\mathbf{H}^\dagger = \mathbf{H}^H (\mathbf{H}\mathbf{H}^H)^{-1}$. When the target Ambisonics order N is high, $(N+1)^2$ can greatly exceed the number of microphones M , and $\mathbf{H}$ can become ill-conditioned due to limited aperture and spatial aliasing. In practice, $\mathbf{H}^\dagger$ is replaced by a regularized inverse that improves numerical conditioning by discarding the effect of small singular values, without resolving the underlying ambiguity [16]. In this regime, the least-squares solution cannot recover spatial detail not supported by the array.

Since the inverse problem is ill-posed, many HOA RIRs $\mathbf{a}$ can explain the same measurements $\mathbf{x}$. We therefore formulate the task as posterior inference and aim to model the conditional distribution $p(\mathbf{a} | \mathbf{x}, \mathbf{h})$ using a data-driven generative model. The posterior can be factorized as: $p(\mathbf{a} | \mathbf{x}, \mathbf{h}) \propto p(\mathbf{a})p(\mathbf{x} | \mathbf{a}, \mathbf{h})$. Because the SHD representation is device independent, this formulation naturally separates two components. The prior $p(\mathbf{a})$ captures the statistics of plausible HOA RIRs and can be learned using a generative model trained with a dataset of HOA RIRs. The likelihood term $p(\mathbf{x} | \mathbf{a}, \mathbf{h})$ is defined by the known forward model in (1), which describes how the microphone measurements are generated from the HOA coefficients for a given array transfer function. At inference time, the two terms are combined: the likelihood enforces consistency with the microphone measurements, while the prior supplies plausible spatial detail that cannot be uniquely determined from the array observations.

III. METHODS

A. Diffusion Model Formulation

We model the distribution of time-domain HOA RIRs $p(\mathbf{a})$ using a generative diffusion model [11]. While recent works proposed generative approaches for RIRs [17]–[21], no generative model has been proposed for HOA RIRs up to date. Diffusion models learn a mapping between a prior distribution (typically a Gaussian) and an empirical data distribution, in our case observed through a dataset of HOA RIRs. This mapping is parameterized by a time variable τ . Generation can be achieved by iteratively traversing the mapping from a high-noise regime toward the data distribution.

Each training example $\mathbf{a} \in \mathbb{R}^{(N+1)^2 \times T}$ consists of a HOA RIR, padded to a fixed T time samples. A continuous set of the noisy marginals $p(\mathbf{a}_\tau)$ is constructed by defining an isotropic diffusion process in the data space, where independent Gaussian noise with identical variance is added to every Ambisonics coefficient and temporal sample. Similar to [22], we define the perturbation process $\mathbf{a}_\tau = \mathbf{a} + \tau \mathbf{z}$, where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Intuitively, increasing τ yields

progressively more corrupted versions of $\mathbf{a}$, and for sufficiently large τ the distribution of $\mathbf{a}_\tau$ approaches an isotropic Gaussian.

Sampling can be formulated by means of the *probability flow ODE*, which transports samples from a Gaussian distribution toward the data distribution:

$$d\mathbf{a}_\tau = -\tau \nabla_{\mathbf{a}_\tau} \log p(\mathbf{a}_\tau) d\tau. \quad (2)$$

This allows generating samples by integrating the ODE down to $\tau \rightarrow 0$. The score function $\nabla_{\mathbf{a}_\tau} \log p(\mathbf{a}_\tau)$ is not available in closed form. Nevertheless, it can be approximated. Assuming a Gaussian perturbation process, the score can be related to the minimum mean-square error (MMSE) denoiser $\mathbb{E}[\mathbf{a}_0 | \mathbf{a}_\tau]$ via [11]:

$$\nabla_{\mathbf{a}_\tau} \log p(\mathbf{a}_\tau) = -(\mathbf{a}_\tau - \mathbb{E}[\mathbf{a}_0 | \mathbf{a}_\tau])/\tau^2. \quad (3)$$

Diffusion models approximate the MMSE estimate using a neural network, i.e., $\mathbb{E}[\mathbf{a}_0 | \mathbf{a}_\tau] \approx \hat{\mathbf{a}}_0(\mathbf{a}_\tau)$. Similar to Karras et al. [22], we parameterize the denoised prediction with a residual model

$$\hat{\mathbf{a}}_0(\mathbf{a}_\tau) = c_{\text{skip}}(\tau) \mathbf{a}_\tau + c_{\text{out}}(\tau) F_\psi(c_{\text{in}}(\tau) \mathbf{a}_\tau, \tau), \quad (4)$$

where F_ψ is a deep neural network, and $c_{\text{skip}}(\tau)$, $c_{\text{out}}(\tau)$, $c_{\text{in}}(\tau)$, and $c_{\text{noise}}(\tau)$ are scalar weighting functions of the noise level τ .

Different to [22], we train F_ψ using conditional flow matching [23], which empirically yields faster optimization in our setting. Specifically, for samples $\mathbf{a} \sim p(\mathbf{a})$, noise $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and τ drawn from a training distribution, we minimize

$$\mathbb{E}_{\mathbf{a}, \mathbf{z}, \tau} [\|F_\psi(c_{\text{in}}(\tau) (\mathbf{a} + \tau \mathbf{z}), \tau) - (\mathbf{z} - \mathbf{a})\|_2^2], \quad (5)$$

This design choice requires setting the preconditioning parameters as $c_{\text{in}}(\tau) = c_{\text{skip}}(\tau) = \frac{1}{\tau+1}$, $c_{\text{out}}(\tau) = \frac{\tau}{\tau+1}$ [24].

B. Hybrid Two-Stage Architecture for HOA RIRs

An important design choice is the architecture of the backbone model F_ψ . In our setting, the model operates on raw Ambisonic RIRs represented with all coefficients stacked along the channel axis. Because this representation is high-dimensional and considering that large-scale datasets are not available, incorporating inductive biases derived from domain knowledge can be advantageous.

Our design is motivated by the distinct structure of RIRs across time. The early part of a RIR is dominated by the direct path and a set of early reflections. Accurately modeling these components is essential to preserve spatial cues, since their relative timing and directional content largely determines the perceived localization. In contrast, the late reverberation is more diffuse and exhibits smoother temporal structure, with frequency-dependent energy decay rates that are naturally captured in the time-frequency domain.

We adopt a hybrid two-stage architecture that combines time-frequency and time-domain processing, summarized in Fig. 1. The first stage processes the input with a two-dimensional network operating on the STFT representation. Here, we adopt a neural network based on NCSN++, following prior work [25], [26]. The inverse STFT is applied at the output to map the intermediate estimate back to the time domain; importantly, this operation is differentiable, enabling end-to-end training. The second stage is designed to refine early reflections. We extract the early segment of both the input noisy signal and the first-stage time-domain output using a fixed mixing time of 70 ms. These early segments are stacked and refined with a time-convolutional U-Net, which is more capable for modeling sparse, transient structure. The final output combines the refined early segment with the late segment produced by the first-stage network.

To enforce rotation equivariance in Ambisonic RIRs, so results remain consistent regardless of orientation, we apply random rotations and their inverses at the input and output of F_ψ . Specifically, we use a Wigner-D rotation matrix $\mathbf{R}_{\phi, \theta}$ where the azimuth ϕ and zenith θ

TABLE I
RESULTS ON FDTD (LEFT) AND TREBLE-10 (RIGHT) (MEAN $\pm$ STD);
LOWER IS BETTER. BEST IN **BOLD**, SECOND BEST IN UNDERLINE.

Method	Internal FDTD		Treble-10	
	EDC $\downarrow$	NPM $\downarrow$	EDC $\downarrow$	NPM $\downarrow$
Linear Encoder	0.049 $\pm$ 0.05	0.999 $\pm$ 0.02	0.106 $\pm$ 0.12	1.042 $\pm$ 0.04
Neural Encoder	0.043 $\pm$ 0.04	1.080 $\pm$ 0.01	0.067 $\pm$ 0.01	1.120 $\pm$ 0.01
Cond. Diffusion	0.026 $\pm$ 0.03	<u>0.938</u> $\pm$ 0.07	0.022 $\pm$ 0.03	0.912 $\pm$ 0.10
Ours	0.018 $\pm$ 0.02	0.874 $\pm$ 0.13	0.019 $\pm$ 0.01	0.803 $\pm$ 0.16
Ours (meas. dist.)	0.025 $\pm$ 0.02	0.967 $\pm$ 0.07	0.026 $\pm$ 0.02	0.849 $\pm$ 0.14
Ours (STFT)	0.021 $\pm$ 0.02	1.065 $\pm$ 0.01	0.047 $\pm$ 0.02	1.083 $\pm$ 0.02
Ours (waveform)	0.022 $\pm$ 0.03	1.012 $\pm$ 0.04	0.018 $\pm$ 0.01	0.950 $\pm$ 0.11

the original work, our task involves encoding room impulse responses (rather than scenes) and targets higher Ambisonics orders (8th/12th), which are beyond the original scope. The third baseline, *Conditional Diffusion*, is a trained to directly approximate $p(\mathbf{a}|\mathbf{x}, \mathbf{h})$ for a specific device. Here, the backbone F_ψ is conditioned on $\mathbf{c} = \mathcal{F}^{-1}(\mathbf{H}^\dagger \mathbf{X}) \in \mathbb{R}^{(N+1)^2 \times T}$, which is concatenated to the input $\mathbf{a}_\tau$, and Classifier-Free Guidance [34] (weight 2) is used to enhance conditioning.

D. Objective Metrics

We evaluate the dissimilarity between ground truth and estimated RIR using two metrics. The Energy Decay Convergence (EDC) [35] measures dissimilarity in late reverberation by comparing the energy decay curves of the two signals. For HOA, EDC is averaged across all Ambisonics coefficients. For early reflections (first 100 ms), we use the Normalized Projection Misalignment (NPM) [36], which quantifies scale-invariant misalignment between the true and estimated signals, also averaged across Ambisonics coefficients.

E. Results

The evaluation results for both datasets are summarized in Table I. Across all metrics and datasets, the proposed method demonstrates clear improvements over the linear encoder baseline, achieving higher accuracy in both early and late reverberation characteristics. The neural encoder, in contrast, consistently underperforms, which we attribute to the method is not capable of providing a reliable estimate at higher orders. The conditional diffusion baseline also yields lower performance compared to our approach. We hypothesize that this is due to the absence of explicit range-space consistency constraints, which are central to the effectiveness of our method.

In ablation studies, replacing the proposed range-projected distance with a measurement distance in the likelihood objective results in a consistent decrease in performance across all evaluation metrics, underscoring the importance of the range-projection constraint. Furthermore, alternative backbone architectures based solely on STFT or waveform processing exhibit a marked reduction in early reflection accuracy, as indicated by the NPM metric results.

F. Perceptual Evaluation

We conducted a listening test to assess the perceptual impact of the proposed method relative to the baselines. We considered two sources of Ambisonics reference RIRs: three examples from the FDTD test set and three internally measured RIRs from an Eigenmike-64 array. The latter were encoded to 7th-order Ambisonics using regularized least-squares [3]. For each reference RIR, we simulated microphone array measurements by applying the Aria Glasses ATF, yielding device-specific signals, and then re-estimated Ambisonics RIRs using the proposed method and the three baselines.

We applied three rotations to the estimated HOA RIRs: no rotation ($\phi = 0^\circ, \theta = 0^\circ$), azimuth-only ($\phi = 45^\circ, \theta = 0^\circ$), and combined azimuth and elevation ($\phi = 90^\circ, \theta = 30^\circ$). Applying the rotations after Ambisonics encoding makes the task more challenging, as the binaural RIRs are evaluated for source directions not directly aligned

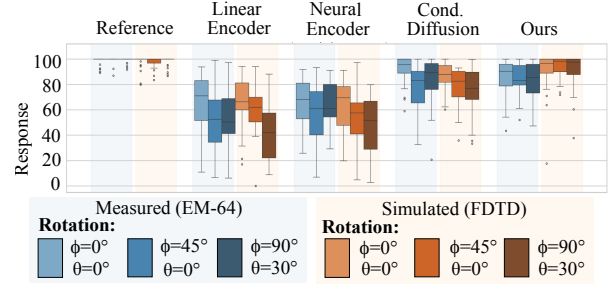


Fig. 2. Boxplot representations of the listening test results.

with the original array configuration. All Ambisonics RIRs (reference and re-estimated) were rendered to binaural signals using a generic HRTF set, and a broadband drum-loop signal was convolved with each binaural RIR to generate the final stimuli.

The listening test followed a multi-stimulus paradigm with an explicit reference. In each trial, listeners were presented with the binaural reference and the corresponding processed stimuli (proposed method and baselines) in randomized order. The reference remained available throughout, and no low-quality anchor was included. Participants rated each processed stimulus on a continuous scale from 0 (very dissimilar to the reference) to 100 (perceptually indistinguishable). Each configuration was repeated twice, conforming 12 trials per experiment. A total of 13 normal-hearing listeners participated.

Figure 2 shows box plots of the similarity ratings. For the simulated FDTD examples, the proposed method clearly outperformed all baselines, with median ratings close to the top of the scale and minimal degradation under rotation, while the baselines showed a marked drop in quality for rotated conditions. For the measured Eigenmike examples, ratings of the proposed method were overall lower due to the distribution mismatch, but the proposed method and the device-specific conditional diffusion baseline achieved similarly high scores and remained clearly above the linear and neural encoders. These results indicate that the proposed device-agnostic diffusion model generalizes to out-of-distribution measured RIRs, providing more faithful binaural reproduction than other approaches.

V. CONCLUSIONS

We have presented a novel method for Ambisonics encoding of RIRs from microphone arrays based on a diffusion-based generative model, trained solely on Ambisonics-domain data. Experiments on simulated data demonstrate that our approach achieves strong performance up to order 12, strongly surpassing the capabilities of a linear encoder. Furthermore, a subjective listening test with binaural renderings of both simulated and measured RIRs shows that our method achieves high perceptual similarity to reference Ambisonics RIRs while remaining robust under rotated listening conditions. Crucially, although the evaluation is limited to a single array, the underlying method is device-agnostic. The trained model is independent of the physical array design and can be applied to any configuration for which the ATF is available, as supported by preliminary internal testing.

Several avenues remain for future research. First, while our current pipeline assumes a highly accurate ATF, incorporating measurement uncertainty would improve robustness to real-world error. Secondly, a substantial distribution mismatch remains between the simulated RIRs used for training and real-world measurements, which can present notable performance limitations. This gap could be addressed either by introducing domain adaptation strategies or by incorporating measured data into training. Finally, while the computationally intensive iterative sampling is already well-suited for offline applications like data generation, future work could focus on enabling faster inference.

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